A PROJECT TITLE

ON

Travel Time Prediction Using Ridge and Lasso Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

C.S.I.T

A.Y.: 2026-27

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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Travel Time Prediction Using Ridge and Lasso Regression

1. Abstract:

Travel time prediction is a crucial component of intelligent transportation systems, enabling efficient route planning, traffic management, and improved commuter experiences. Accurate prediction of travel time helps reduce traffic congestion, minimizes fuel consumption, and supports informed decision-making for both travelers and transportation authorities. However, predicting travel time is a challenging task due to the influence of multiple dynamic factors such as traffic density, weather conditions, road infrastructure, time of day, and unexpected incidents. This study presents a travel time prediction model using **Ridge Regression** and **Lasso Regression**, two widely used regularization techniques that improve the performance of linear regression models by reducing overfitting and handling multicollinearity among input variables.

The proposed approach begins with data preprocessing, including handling missing values, feature selection, normalization, and splitting the dataset into training and testing sets. Ridge Regression applies L2 regularization to shrink coefficient values while retaining all features, making it suitable for datasets with correlated variables. In contrast, Lasso Regression employs L1 regularization, which not only minimizes prediction error but also performs automatic feature selection by assigning zero coefficients to less significant variables. The predictive performance of both models is evaluated using standard regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2 score).

The comparative analysis highlights the strengths and limitations of each regression technique in predicting travel time under varying traffic conditions. Results demonstrate that Ridge Regression provides stable predictions in the presence of multicollinearity, while Lasso Regression enhances model interpretability by identifying the most influential features affecting travel time. The findings indicate that both methods offer reliable and computationally efficient solutions for travel time prediction, making them suitable for real-time transportation applications. This research contributes to the development of intelligent traffic prediction systems that can support smarter urban mobility, optimize route planning, and improve the overall efficiency of modern transportation networks.

2. INTRODUCTION:

Travel time prediction is an important application of machine learning that helps estimate the duration of a journey more accurately. The prediction depends on various factors such as distance, traffic conditions, weather, day of the week, and departure time. This project develops a web-based application that predicts travel time between a selected source and destination. The system uses route information along with real-time traffic and weather data as input features. Ridge Regression and Lasso Regression models are implemented to generate accurate travel time predictions. The application also displays the estimated arrival time (ETA) and route on an interactive map. This project demonstrates the practical use of machine learning in improving travel planning and transportation efficiency.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11 or Ubuntu Linux
- Programming Language: Python 3.x
- Frontend: HTML5, CSS3, JavaScript
- Backend Framework: Flask
- Machine Learning Library: Scikit-learn
- Data Processing: Pandas and NumPy
- Visualization: Matplotlib
- APIs: Open Route Service/Google Maps API and Open Weather API
- Development Tool: Visual Studio Code
- Web Browser: Google Chrome or Microsoft Edge

Signature of the Guide **Course Instructor**